

ALGAETOPIA

A Serious Game Exploring Future Marine Symbiotic Societies in 2060

Concept & Vision

ALGAETOPIA is an immersive **serious game** designed for **entities** and **institutions** such as natural resource departments, civil affairs agencies, and construction companies. By simulating the development process of a **future community in 2060**, it assists in the planning and formulation of current **sustainable development strategies**. The game focuses on the role of **algae** in an aging society and explores the potential for symbiotic coexistence between humans and the marine environment.

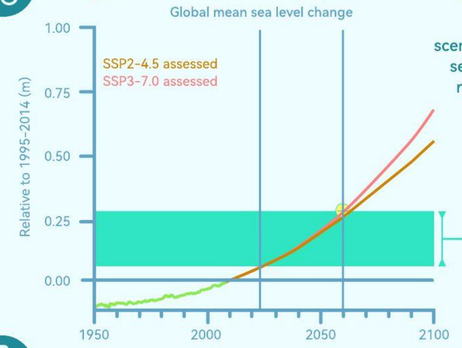
Project Date: 2024
Team Project (5 members) | My Role: Project Lead & Technical Director
Contribution: Modeling (90%), FX, Texturing, Rendering, UI Design (10%), UE5 Development, & Curation.

The 2060 Scenario: Environment & People

situated within the global context, with a specific focus on **China's Pearl River Delta region**.

Challenges & Opportunities

Rising sea levels driven by global warming



Under the SSP3-7.0 scenario, the global mean sea level is projected to rise by **0.26 meters** by 2060.

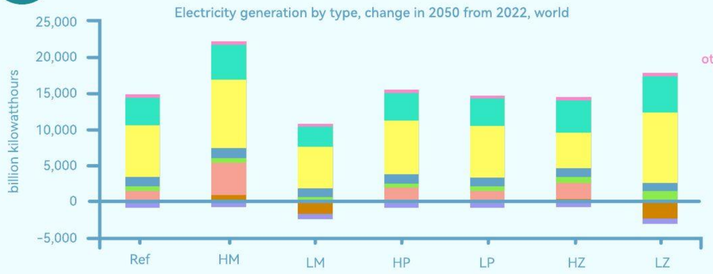
Data Source: IPCC AR6, Working Group I, 2021.
Note: SSP=Shared Socioeconomic Pathways.

Maritime City Concept



Although maritime cities sound like a concept straight out of science fiction, in a future marked by rising sea levels and energy shortages, migrating to the ocean could indeed be a more ideal way of survival. After all, the ocean covers 71% of the Earth's surface.

Growing energy demand and rising supply risks



Data source: U.S. Energy Information Administration, International Energy Outlook 2023 (IEO2023)
Note: Ref=Reference case; HM=High Economic Growth; LM=Low Economic Growth; HP=High Oil Price; LP=Low Oil Price; HZ=High Zero-Carbon Technology Cost; LZ=Low Zero-Carbon Technology Cost.

Projections indicate that by 2050, **Variable Renewable Energy** (VRE) will dominate the global energy mix. However, the intermittency of wind and solar power may destabilize the grid, posing a potential **energy supply risk** amid growing demand for machinery and automation.

ALGAE as Living Infrastructure

Input

- **CO2 Emissions** From Flue Gas & Respiration
- **Wastewater** (Nitrogen & Phosphorus Source)

The Bio-Engine



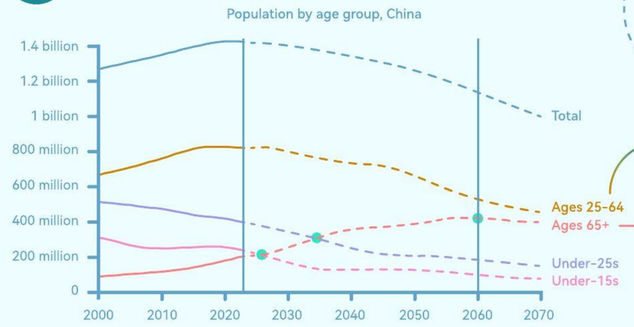


Output

- **Food & Feed** Single Cell Protein & Functional Food
- **Bio-Energy** Single Cell Protein & Functional Food
- **Eco-Service** O2 Release & Water Purification

Adapted from: Sepulveda, C., et al. (2019). "Comparative evaluation of microalgae strains for CO2 capture purposes." Journal of CO2 Utilization, 30, 158–167.

Challenges of population inversion and passive elderly care.



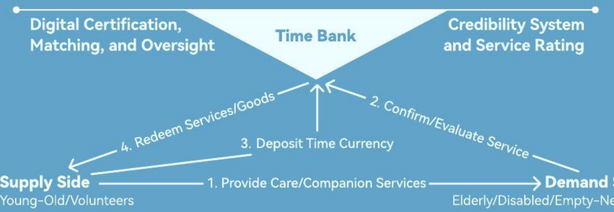
Data source: UN, World Population Prospects (2024)

While aging is a gradual process globally, China's demographic data is expected to undergo a dramatic reversal around 2060. The country's **core working-age population** (ages 25–64) will peak after 2025 and then gradually **decline**, with the contraction accelerating after 2045.

By 2060, the population aged 65 and above is projected to nearly match the size of the working-age population.

Community-Based Mutual Support in Elderly Care

Take the **Time Bank model** as an example



Adapted from: Population Research Institute of Peking University & Chinese Red Cross Foundation. (2021). China Time Bank Development Research Report.

Future Persona



1988 | Mr. Wang, aged 22

Living in the early period of reform and opening up in China, amid the industrial boom of the "three supplies and one compensation" (processing with supplied materials, samples, or parts, and compensatory trade) in the Pearl River Delta region.

- Material ownership > Right to experiences
- Linear Life Path
- Collectivism



2024 | Mr. Wang, aged 58

Facing the backdrop of the onset of China's deep aging, and amidst the legacy of the one-child policy, as the **2** in the 4-2-1 family structure (supporting four elderly parents above and raising one child below).

- Anxiety over Filial Dependence
- Digital Divide
- Passive Aging



2024 | Ms. Liu, aged 22

Living in a post-industrial era and an atomized society, marked by the coexistence of involution and lying flat.

- Right to experiences > Material ownership
- Eco-Anxiety
- Atomization



2060 | Mrs. Liu, aged 58

Facing the normalization of climate crisis and the era of human-machine symbiosis. **Productive Aging** is likely to become mainstream, where the elderly are no longer mere consumers of resources but creators of social value. The resident model in the game is constructed based on this logic.

Design References for Game Mechanics:

Projected population profiles for 2060 enable professional civil affairs agencies to observe **behavioral data** of in-game residents in areas such as intergenerational mutual aid, productive contributions, and community resilience.

- Health management > Mere treatment

Residents are accustomed to monitoring of various metrics via **implantable or wearable devices**. Medical facilities in the game are designed to focus more on **long-term** health management and **prevention** rather than treatment of illness.

In-game event mechanisms are based on algorithmically matched interest or skill groups (e.g., microalgae production teams), rather than family units.

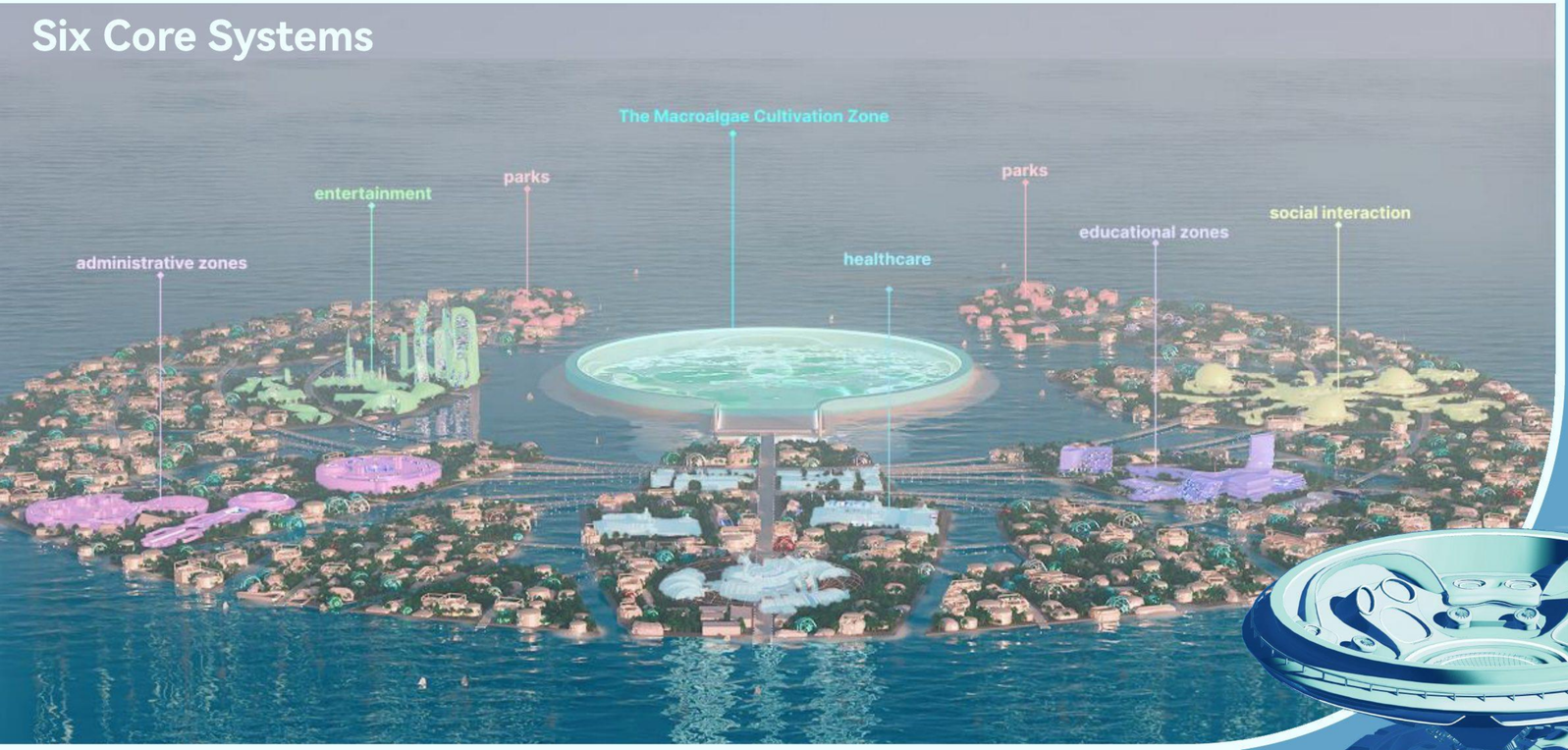
- Elective Kinship
- Productive Aging

Players should create **meaningful work opportunities** or **educational facilities** during the construction phase; otherwise, resident satisfaction will decline.



Spatial Morphology & Zoning

Six Core Systems



Research and Development Center
It serves as the in-game shop

Committed to advancing novel algae-based materials, aquatic construction technologies, and pharmaceuticals, while also overseeing the recruitment and training of community talent.

Energy System
Combines diverse clean energy technologies

Such as algae-based power generation, tidal energy, and solar power. These solutions are both eco-friendly and highly promising for mitigating future global energy crises, ensuring a stable and sustainable energy supply for the community.

Waste Management System
It integrates with the Energy System

Handles the clean and harmless treatment of marine city waste, converting it efficiently into clean energy. This system is crucial for environmental protection, resource recycling, and fostering ecological balance in the community.

Production System
Such as the Macroalgae Cultivation Zone

Includes essential processes like seawater desalination, algae cultivation, and algae-based material production. These activities are vital for community self-sufficiency and sustainable development, underpinning the community's economic cycle.

Living System
Such as the administrative zones

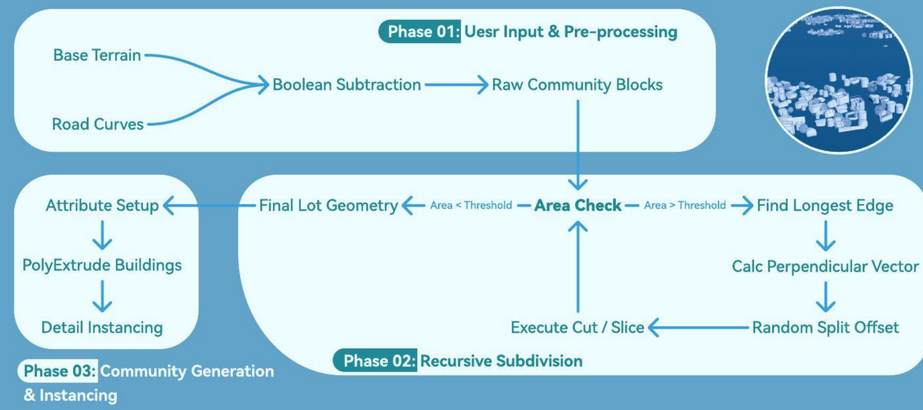
Covers housing, healthcare, social entertainment, and senior education, delivering comprehensive living services and support for residents. This integrated system is essential for improving quality of life, strengthening community cohesion, and promoting social harmony.

Transportation System
Combines land and water transportation

Offering convenient and efficient travel for residents. Optimizing this system significantly enhances community operations, boosts life satisfaction, and strengthens internal and external connectivity.

Recursive Binary Subdivision

The community layout is generated using a recursive splitting algorithm that mimics organic urban growth.





Production System

Photic Macroalgae Canopy
Positioned at the uppermost tier of the production system, this zone utilizes natural light and seawater flux to facilitate the large-scale cultivation of Macroalgae (e.g., giant kelp and Ulva prolifera). It yields basic carbohydrates at the underlying Processing & Synthesis Hub while maintaining the community's ecological equilibrium.

Processing & Synthesis Hub
Converts wet algal biomass into viable energy, materials, and food products.

High-Density Microalgae Zone
Employs closed tubular **photobioreactors** (PBRs) for the cultivation of Microalgae such as Spirulina and Nannochloropsis. Specifically, this zone modulates the light spectrum and nutrient solution ratios to induce the targeted accumulation of specific metabolites.

Active Submersion Flagella
Drawing inspiration from the locomotory mechanisms of flagellated algae, this system actively retracts floating structures to safe underwater depths during extreme weather events, such as typhoons. In Algaetopia, this biomimetic flagella design is implemented across all architectural typologies, extending beyond the Production System.

Gameplay Mechanisms & Interaction

Simulation Gameplay is the core mechanics of ALGAETOPIA, wherein players are tasked with constructing and overseeing a self-sustaining community predominantly inhabited by elderly residents, while exploring the viability of its sustainable operation.

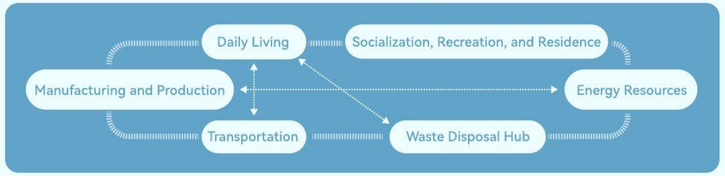


Game start → **Guided Tutorial** → **Game objective: To enable this community to operate sustainably**

- **Step1:** Understanding Basic Natural and Cultural Context
- **Step2:** Comprehending Fundamental Modules
- **Step3:** Constructing a Sustaining Community
- **Step4:** Observing Internal Community Dynamics
- **Step5:** Generating Analytical Reports

Six Core Systems





Initiate a new construction cycle.

Material Prosperity | Infrastructure | Social Welfare | Social Ambiance | Cultural Development

New players will be introduced to the environmental and social context of ALGAETOPIA, as well as the **Six Core Systems** essential for establishing a community, through the **Guided Tutorial**. This phase is designed to equip users swiftly with the foundational knowledge required to comprehend and master the key elements of constructing an in-game geriatric community.

Example: Construction in the Game

Upon completion of the instructional content, players will commence the formal construction of their community. In this phase, the player must comprehensively consider these multiple **real-world factors** to create a feasible and efficiently functioning algae-based retirement community:

- Economic
- Population
- Healthcare
- Resident Needs
- Satisfaction
- Industrial Structure

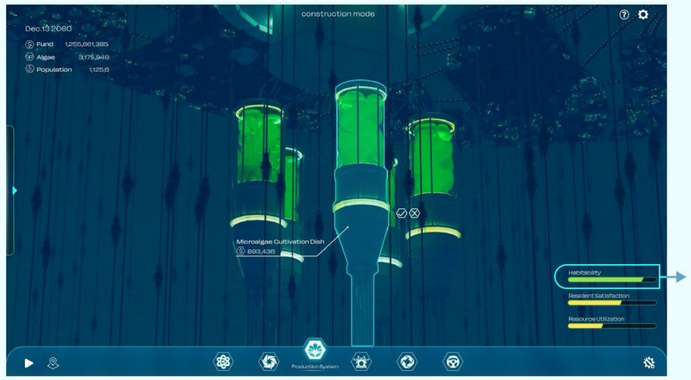
For example, the player is currently building an energy platform, which is a component of the Energy System capable of supplying the necessary power to the Production System.



When the player completes the construction of this platform and places it within the community, although it may result in a decrease in the player's reserve **Funds**, it will simultaneously significantly enhance the efficiency of **Resource Utilization**.

1 → 2





Based on observations of the resource utilization warnings in the lower right corner of the game interface, the player might have decided to add more energy units precisely out of consideration for optimizing Resource Utilization.

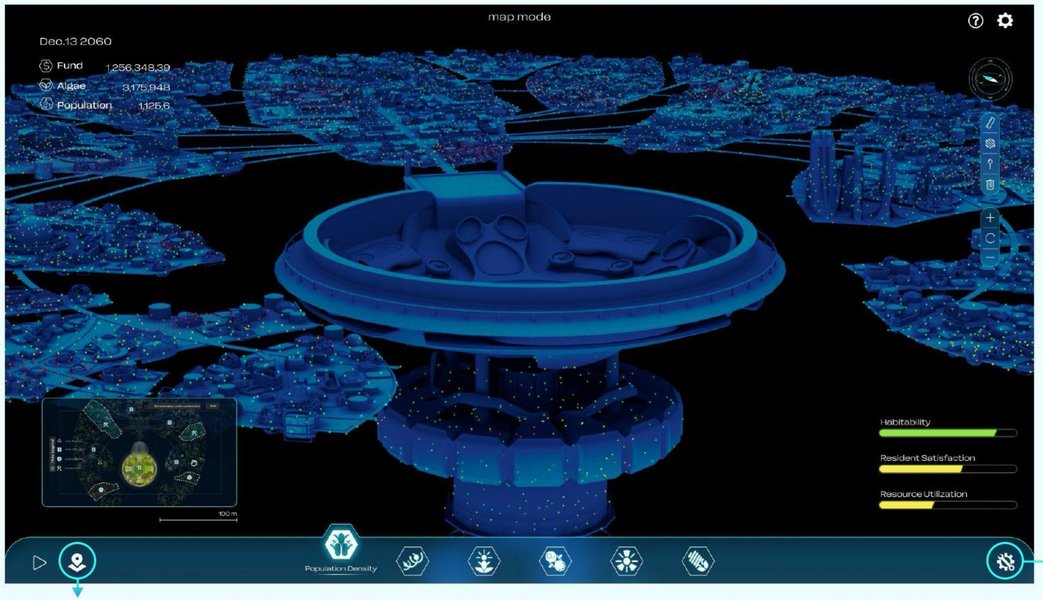
With the continuous enhancement of energy supply, players are enabled to further refine and elaborate the specific components of the Production System.

For instance, the introduction of photobioreactors by the player significantly augments financial and algal output, playing a pivotal role in enhancing the overall **Habitability** of the community and elevating **Resident Satisfaction**.

Example: Construction in the Game ...

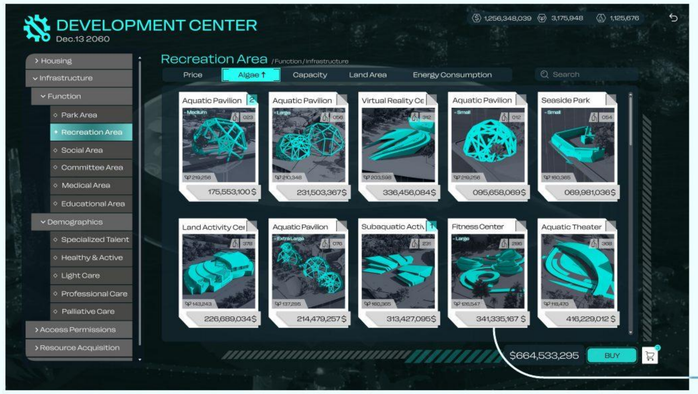
Upon the initial completion of the algae-based mutual aid retirement community, each functional sector commences autonomous operation, adhering to its designated responsibilities.

3



In this stage, players can closely monitor the sustainability of each system component, or pause the simulation via the **Map Mode** to identify potential unforeseen issues during operation.

4



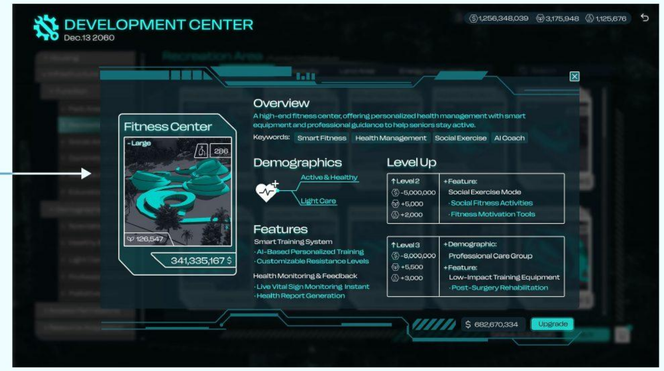
With increased community funds, algal reserves, and population capacity, players can now unlock the **Development Center** to research and develop new building units.

For instance, players might want to develop entertainment areas, which can significantly enhance the community's Habitability index and increase Resident Satisfaction. Consequently, researching and constructing these facilities requires a certain **population capacity** and consumes specific resources.

... and Developing New Building Units

Clicking the image opens a popup displaying detailed information about the building, including its overview, target demographics, features, and upgrade options.

5



The Data Report

Upon the completion of each construction cycle,the game system automatically generates an exhaustive Community Status Report.



To ensure a balanced distribution of information density within the report interface, metrics across each domain have been further categorized. Taking Social Welfare as an example, its metrics are divided into three primary categories: **Population Data, Medical Resources,** and **Healthcare Institutions**. Within the Population Data category, for instance, players can access details on the **Basic population situation**...



This report encompasses **five key areas**: material prosperity, infrastructure, social welfare, social ambiance, and cultural development.

Progression & Feedback

...or switch to other categories such as **Medical Resources**.



Technical Breakdown

Procedural City in Houdini

Input Block → Recursive Binary Subdivision → Final City (Optimized)



Optimization: Packed Geometry

Input Raw Geometry



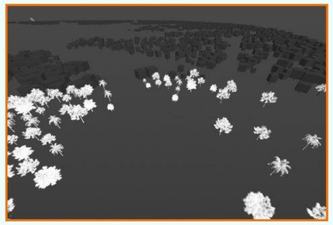
Points 1,630,448 Primitives 1,022,321 Vertices 4,089,284

Primitive Types

Polygon 1,022,321

PACK

Output Packed Array



Points 26,438 Primitives 26,438

Vertices 26,438

Primitive Types

Packed Geometry 26,438

Scatter Points

Attribute Wrangle

@orient, @pscale

s@unreal_instance

Copy to Points

Pack & Instance

Input Geometry

High-Poly Mesh

Transform Matrix	**Pointer Ref**
4x4 Intrinsic Attribute	Geometry stored once in RAM

Houdini Engine API

Data Marshaling & Translation

Pipeline between Houdini and UE5

Houdini Engine & PDG Integration

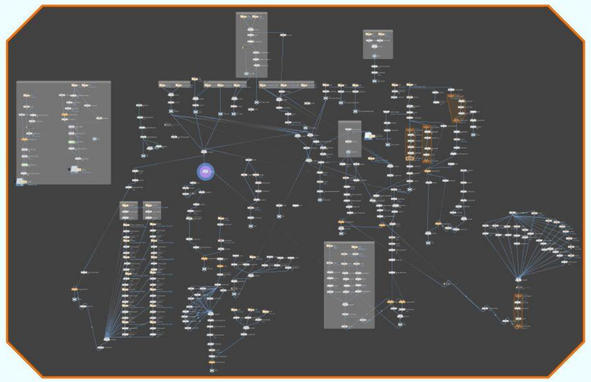
UE5 Input

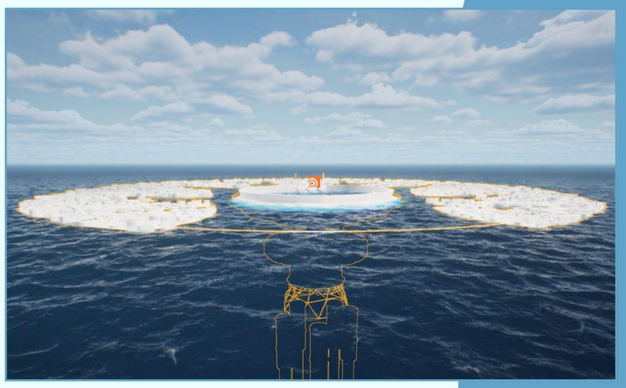


Data Stream

Sea Level and Geographic Coordinates

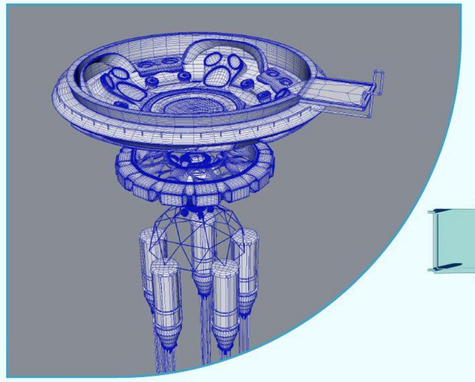
HDA Asset → Hierarchical Instanced Static Mesh → UE5 Output

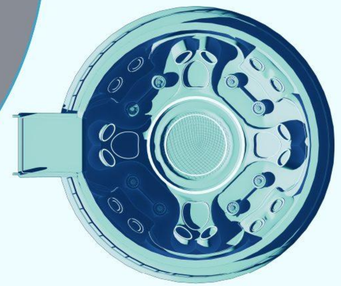


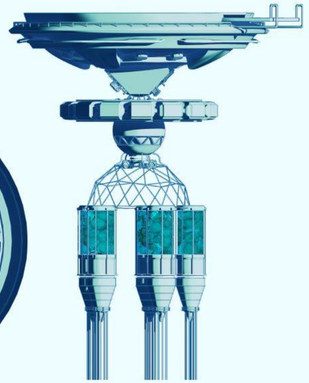


Houdini Attribute (Point/Prim)	UE5 Function (Target)
s@unreal_instance	Instanced Static Mesh
s@unreal_material	Assign Materials
f@unreal_foliage	Foliage Type
s@unreal_bake_folder	Asset Directory

High-Poly Mesh in Maya

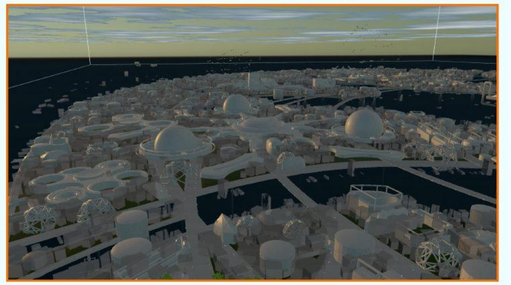






Offline rendering in Karma

Houdini's USD-based Renderer





and Behind the Scenes

Public Exhibition at GOSIM 2025







Selected to present the Algaetopia project at the **Global Open-Source Innovation Meetup**.